

CSC1029 OOProgramming

Lecture 9 – Case Study – Student Register





What's to come in today's lecture?

- Student Register Application
- Introduction to Class Data

A Menu-Driven Application

Develop a program called 'StudentRegister' which uses the Menu class to provide the options listed below:

The application should

- Use an array to hold Student instances
- Continue execution until user quits.

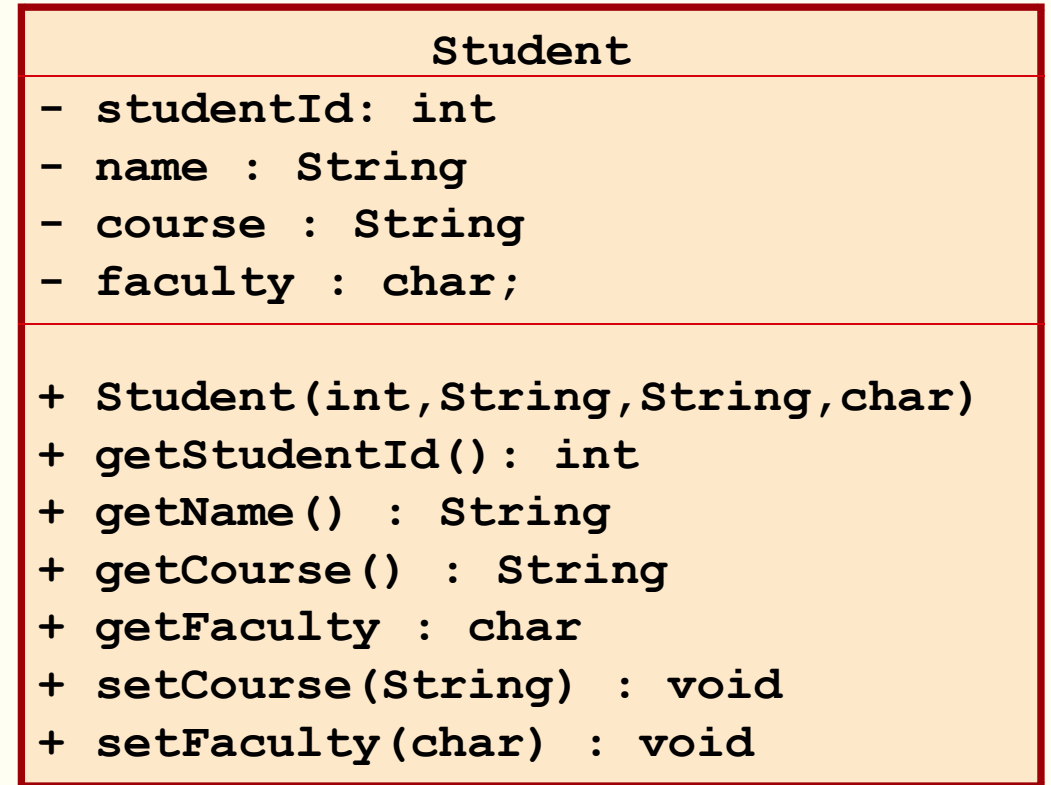
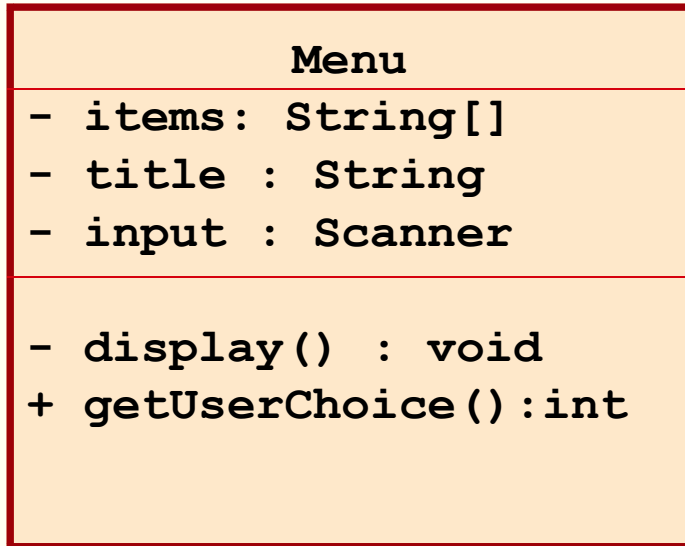
Student Register

+++++

- 1. Add New Student**
- 2. List All Students**
- 3. Quit**

Enter Selection:

UML Specification



StudentRegister

Java Implementation – The Student Class

```
public class Student {  
    private int studentId;  
    private String name;  
    private String course;  
    private char faculty;  
  
    public Student(int id, String sname,  
                  String scourse, char fac) {  
        this.studentId = id;  
        this.name = sname;  
        this.course = scourse;  
        this.faculty = fac;  
    }  
    ...  
}
```

Instance Data

Each instance
(object) will
have separate
values for these

s1

studentId: 1234
name: "Homer"
course: "Computer Science"
faculty: 'S'

s2

studentId: 1235
name: "Marge"
course: "BIT"
faculty: 'S'

s3

studentId: 1236
name: "Lisa"
course: "Software Eng."
faculty: 'S'

s4

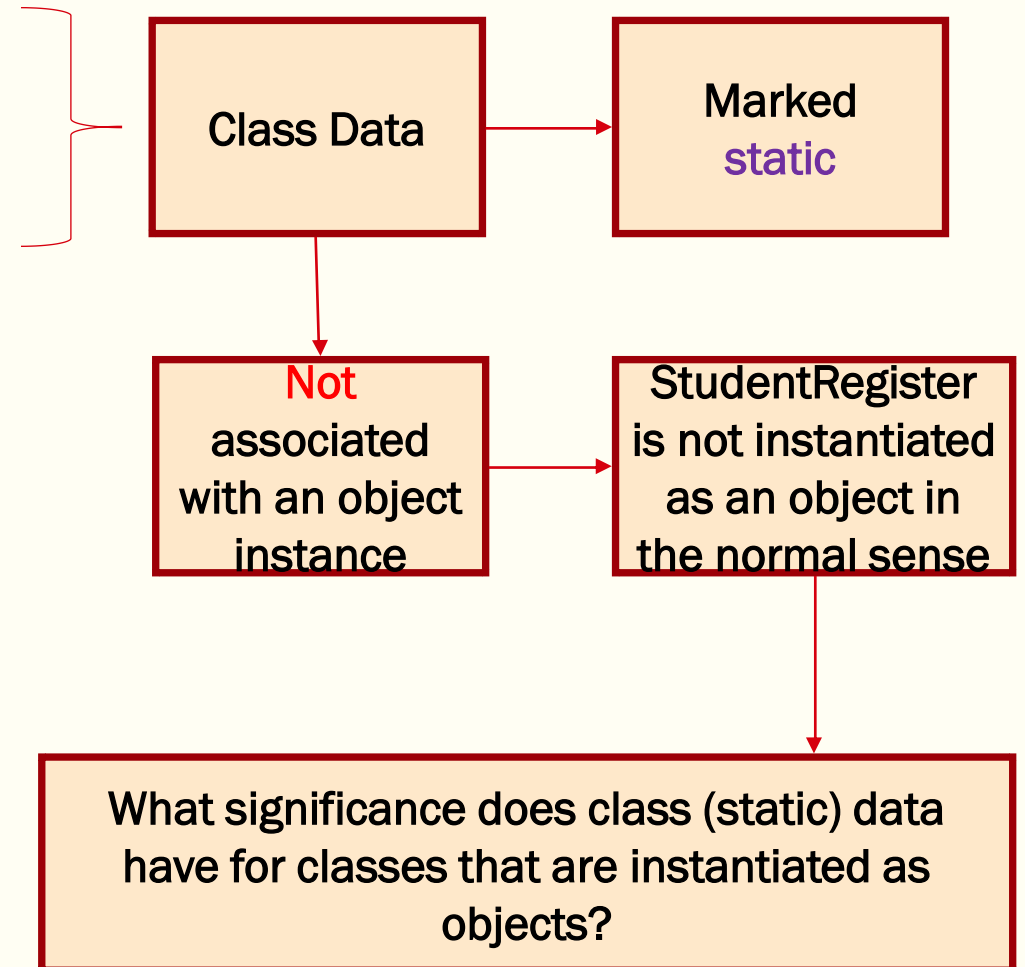
studentId: 1237
name: "Bart"
course: "CIT"
faculty: 'S'

StudentRegister.java

Java Implementation – Application Class

```
public class StudentRegister {  
  
    static final int MAX = 2;  
    static Student data[] = new Student[MAX];  
    static int count = 0;  
  
    public static void main(String[] args) {  
        ...  
    }  
  
    ...  
}
```

StudentRegisterV2.java



Java Implementation – The Student Class ⁽¹⁾

```
public class Student {  
    private int studentId;  
    private String name;  
    private String course;  
    private char faculty;  
  
    private static int nextId = 1234;  
  
    public Student(int id, String sname,  
                  String scourse, char fac) {  
        this.studentId = id;  
        this.name = sname;  
        this.course = scourse;  
        this.faculty = fac;  
    }  
}
```

Instance Data

Class Data

Only one copy
shared across
all instances

s1

studentId: 1234
name: "Homer"
course: "Computer Science"
faculty: 'S'

s2

studentId: 1235
name: "Marge"
course: "BIT"
faculty: 'S'

s3

studentId: 1236
name: "Lisa"
course: "Software Eng."
faculty: 'S'

s4

studentId: 1237
name: "Bart"
course: "CIT"
faculty: 'S'

StudentV2.java

Java Implementation – The Student Class ⁽²⁾

```
public class Student {  
    private int studentId;  
    private String name;  
    private String course;  
    private char faculty;  
  
    private static int nextId = 1234;  
  
    public Student(String sname,  
                   String scourse, char fac) {  
        this.studentId = nextId;  
        nextId++;  
        this.name = sname;  
        this.course = scourse;  
        this.faculty = fac;  
    }  
}
```

In this example – we can use `nextId` to help assign `studentId` – automatically incrementing within the constructor.

```
Student s1 = new Student("Homer Simpson",  
                          "Computer Science", 'S');  
Student s2 = new Student("Bart Simpson",  
                          "Software Engineering", 'S');
```

Number: 1234
Name: Homer Simpson
Course: Computer Science
Faculty: S

Number: 1235
Name: Bart Simpson
Course: Software Engineering
Faculty: S



What's Next?

Class Variables & Methods – the Detail

A Bank Account Example

